

(Time: 2 hours)

[Total Marks: 60]

- N. B.: (1) All questions are compulsory.
 (2) Make suitable assumptions wherever necessary and state the assumptions made.
 (3) Answers to the same question must be written together.
 (4) Numbers to the right indicate marks.
 (5) Draw neat labeled diagrams wherever necessary.
 (6) Use of Non-programmable calculator is allowed.

1. **Attempt any two of the following:** 12
 - a. Explain the request and response communication model.
 - b. Explain the push-pull communication model.
 - c. Explain the application of IoT in smart retail.
 - d. How is IoT applied in smart cities? Explain.

2. **Attempt any two of the following:** 12
 - a. Explain the cloud based architecture of IoT.
 - b. What is NFC? Explain. What are wireless sensor networks?
 - c. What is Raspberry Pi? What are the hardware requirements for Raspberry Pi? Why is it used?
 - d. What is Internet Gateway Device? State the five stages when Internet Gateway Devices are used.

3. **Attempt any two of the following:** 12
 - a. What are different types of hypervisors? Explain the benefits and limitations of each.
 - b. Explain single root and multi root input output virtualization.
 - c. What are the different data security issues in IoT?
 - d. Discuss the IoT ecosystem connectivity.

4. **Attempt any two of the following:** 12
 - a. Without network connectivity, IoT adoption will have no relevance. Discuss this assertion supporting it with relevant examples.
 - b. How is IoT used in politics and military? Explain.
 - c. What is the role of cloud in IoT? Discuss.
 - d. How IoT can be used for (i) Remote operations in construction (ii) Logistics supply in military (iii) Remote operations in construction?

5. **Attempt any two of the following:** 12
 - a. Enumerate the differences between smart cities and non-smart cities.
 - b. Is there any connection between smart city and resilient city? Discuss.
 - c. Discuss the Copenhagen zero carbon emission agenda and district of Songdo new resident integration.
 - d. Describe a futuristic situation where you will use IoT to provide solution to a problem (either an old or emerging problem) within your community or country.